5. Vertical line

It is a line indicated by a freely suspended plumb bob. It is a line along the direction of gravity.

6. Vertical plane

It is a plane containing a vertical line at a point.

7. Datum surface

It is any level surface taken as a reference for the determination of elevation of various points. The datum commonly used is Mean Sea Level (MSL).

8. Mean Sea Level (MSL)

It is a level surface which represents the average sea water level.

9. Reduced level

The vertical distance above (or) below the datum is known as elevation. It's level is called reduced level.

3.3.3 Benchmark and its types

Benchmark is defined as a fixed reference point of known elevation. It is denoted by BM

The types of benchmarks are:

1. Great Trignometrical Survey Benchmark (GTS)
2. Permanent Benchmark
3. Temporary Benchmark
4. Arbitrary Benchmark

3.3.4 Instruments used in leveling

The instruments used in leveling are

i. Level

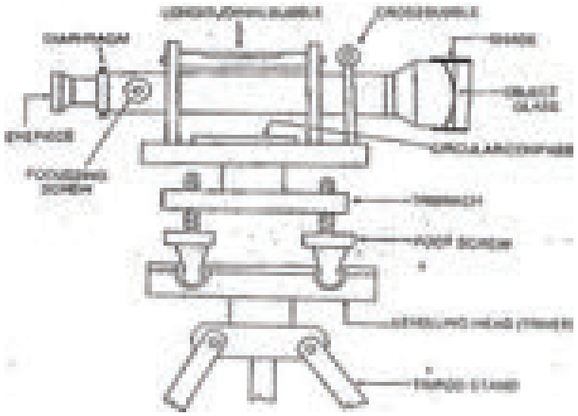
ii. Leveling staff

3.3.5 Different types of levels

The following are the various types of levels

1. Dumpy level
2. Tilting level
3. Quick setting level
4. Wye or Y level
5. Reversible level
6. Automatic level
7. Laser level

3.3.6 Setting up instrument



DUMPY LEVEL

(i) Adjusting of a level

The types of adjustments to be done in a level are

1. Temporary adjustment
2. Permanent adjustment

1. Temporary adjustment

The operations involved in temporary adjustments are

a. Setting up the level

b. Levelling up

c. Elimination of parallax

a. Setting up the level

i. Fixing the instruments on to the tripod.

ii. Approximately leveling the instrument by leg adjustment.

b. Levelling up

The purpose of leveling up is to make the vertical axis truly vertical.

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